Name: _____ Class and student number: _____

Digestion of lipids and fair comparisons

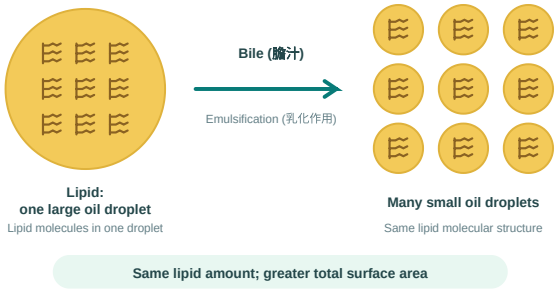
Read A from the start and keep it available. Study the explanations and examples, complete B, then receive C.

Lipids in our food have large molecular structures. They cannot pass directly through the cells lining the digestive tract (消化道) to be absorbed.

X: bile (膽汁) and physical digestion (物理消化)

X is bile (膽汁), made in the liver and stored in the gall bladder (膽囊). It contains no digestive enzymes. Emulsification (乳化作用) spreads large oil droplets into small droplets without changing lipid molecules. For the same total amount of lipid, total surface area increases, giving lipase (脂肪酶) more surface to act on.

01 • Physical digestion (物理消化): emulsification (乳化作用), with no change in molecular structure



Only the size of the oil droplets and how spread out they are change; lipid molecules are not broken down. Small droplets increase the total surface area. The diagram does not show the real sizes.

Y: lipase (脂肪酶) and chemical digestion (化學消化)

Y is lipase (脂肪酶), which speeds up lipid hydrolysis and changes chemical bonds. This course uses a complete-hydrolysis model: triglyceride (甘油三酯) + $3H_2O$ → glycerol (甘油) + 3 fatty acids (脂肪酸). The enzyme is a catalyst and is not used up.

High temperature, denaturation and fair comparison

Lipase (脂肪酶) is a protein. Boiling can cause denaturation and loss of function; cooling does not restore its action in this model. Cool boiled liquids to the common 37°C before comparison, so reaction temperature does not become another difference.

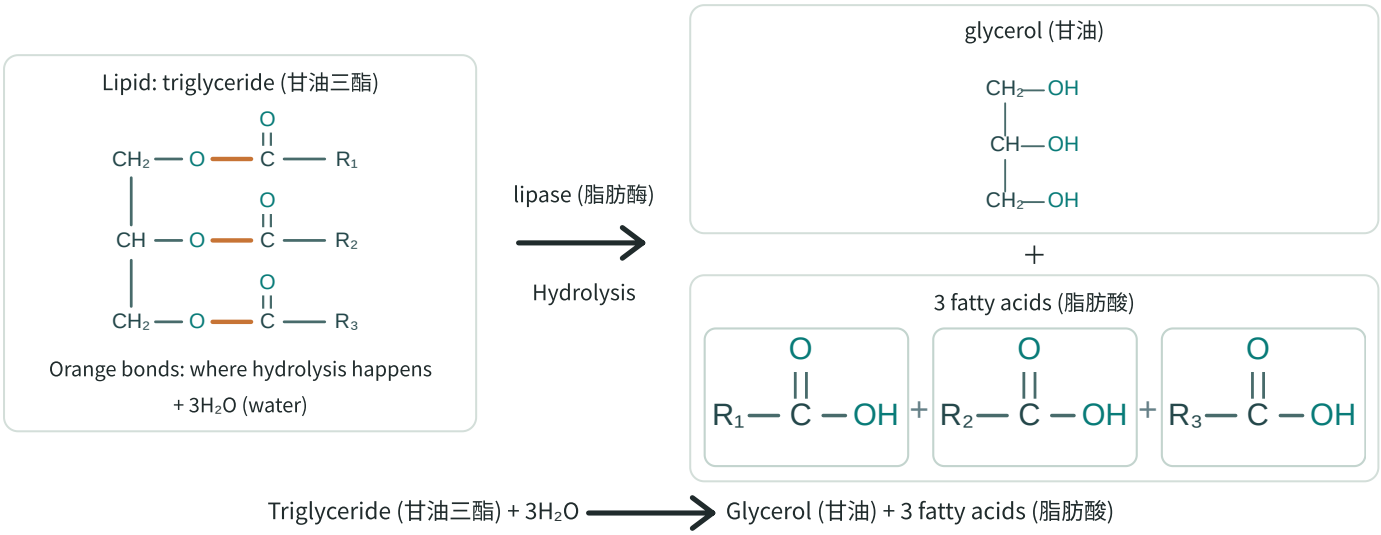
Appearance and product evidence are different

Oil layers, cloudiness and clarity are set teaching clues. They alone cannot prove glycerol (甘油) or fatty acids (脂肪酸) formed, or measure digestion speed. Real experiments need chemical tests and repeated tests.

The total activity time is 15–20 minutes. Follow the teacher’ s time limit for reading, practice, checking and correcting.

Diagrams, methods and worked examples

02 • Chemical digestion (化學消化): hydrolysis forms different molecules



R_1, R_2 and R_3 show the hydrocarbon chains of fatty acids (脂肪酸). Hydrolysis changes chemical bonds to form glycerol (甘油) and fatty acids (脂肪酸). Lipase (脂肪酶) is a catalyst and is not used up in the reaction. This diagram uses the simple complete-hydrolysis model commonly taught in this course.

The four specified sets and fixed conditions

Each set starts with the same amounts of oil and water, receives 2 mL in total, reacts at 37°C, and uses equal observation time and shaking. Recipes: control, 2 mL water; X, 2 mL X; Y, 2 mL Y; X + Y, 1 mL each. Use the researcher’ s common starting amounts, timing and shaking details; do not invent values missing from the model.

How to classify variables

The independent variable is the factor changed on purpose; the dependent variable is the result seen or measured; controlled variables are kept the same. Start with the comparison question, then find the one factor that changes.

Choose the control for the question

A control is a baseline that helps rule out other causes. It need not be free of every factor or reagent. Choose it for the actual question: adding liquid, heating, or another treatment.

Worked example: a testable prediction

“With other conditions equal, adding 2 mL X will spread large oil droplets into small droplets.” Reason: X is bile (膽汁), which can emulsify lipids (乳化脂質). Compare water with X and observe appearances. This is a worked example, not your original hypothesis. A prediction that can be observed and compared is testable; whether data support it is a separate judgement.

Worked example: classification and observation

If the question compares shaking methods, shaking is independent, the observed result is dependent, and the juice combination and other conditions stay the same. This teaches the method; it is not B’ s main-comparison answer. “Particles are at the bottom” is an observation; “they have chemically broken down” needs other evidence.

Appearance is a teaching clue: clarity alone does not prove glycerol (甘油) and fatty acids (脂肪酸) formed or measure speed. X + Y uses 1 mL each, while a single-juice set uses 2 mL. The different individual amounts cannot prove a measured combined effect or faster rate.

Guided practice: observation and judgement

Circle one answer in each set of brackets; use boxes for multiple choices. Each written answer has two lines.

Initial oil + water



B1.1 Describe only the oil–water test tube before any digestive juice is added. What layers and boundary can you see?

Describe what you can see. Keep explanations such as “digested” separate.

B1.2 Evaluate this given prediction and reason. You do not need to invent an “original hypothesis”.

“With the other conditions kept the same, adding 2 mL Y will spread large oil droplets into small droplets, because all digestive enzymes emulsify lipids (乳化脂質).”

Judge separately whether observations can test the prediction and whether the reason fits the concepts. Then explain the problem with the reason.

Can observations test it? （Yes ／ No）

Does its reason fit the concepts? （Yes ／ No）

Brief reason / explanation:

B1.3 The main comparison changes the combination of digestive juices. Classify the factors as independent, dependent or controlled variables.

Use the method in A. More than one factor may belong to a group.

Independent variable (the factor changed on purpose)	Dependent variable (the result you measure or observe)	Controlled variables (the factors kept the same)
☐ Total volume of oil and water	☐ Total volume of oil and water	☐ Total volume of oil and water
☐ Reaction temperature	☐ Reaction temperature	☐ Reaction temperature
☐ Reaction time	☐ Reaction time	☐ Reaction time
☐ Total volume of liquid added	☐ Total volume of liquid added	☐ Total volume of liquid added
☐ Method of shaking	☐ Method of shaking	☐ Method of shaking
☐ Appearance of the mixture	☐ Appearance of the mixture	☐ Appearance of the mixture
☐ Combination of digestive juices	☐ Combination of digestive juices	☐ Combination of digestive juices

B1.4 Choose all the conditions needed for a fair comparison.

An assumption is a condition taken as true when explaining results. The procedure must also follow the fixed conditions provided.

- ☐ Observe all setups after the same reaction time.
- ☐ The X + Y group can have more liquid in total to compare it with the other groups.
- ☐ Use the same starting amounts of oil and water and the same total volume of added liquid in each setup.
- ☐ Use the same method of shaking for every setup.

Guided practice: controls, labels and fairness

Common comparison conditions

Each set starts with the same amounts of oil and water, receives 2 mL in total, reacts at 37°C, and uses equal observation time and shaking. Recipes: control, 2 mL water; X, 2 mL X; Y, 2 mL Y; X + Y, 1 mL each. Use the researcher’ s common starting amounts, timing and shaking details; do not invent values missing from the model.

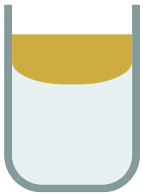
B2.1 To compare adding digestive juice with adding only water, which of the four sets is the baseline? Why keep the total volume added the same?

Choose the control for this specific question. A change in another condition must not be mistaken for the effect of the juice.

B2.2 The four basic setups are already specified. Use the recipes in A to complete each label and added volume; no new apparatus design is needed.

Each setup receives 2 mL in total and starts with the same amounts of oil and water.

Setup 1



Setup 2



Setup 3



Setup 4



B2.3 Someone adds 2 mL water to the control but 2 mL X and 2 mL Y to the combined set, with other conditions equal. Is this fair? Identify the problem, fix the two volumes and explain why.

First find the total volume added to each set. Then check whether only one factor changed.

Complete and keep the first answers in B. After receiving C, correct in another colour without erasing the original.

Independent application: four main sets

B3.1 Independent application: observe the four static images and record their appearances.

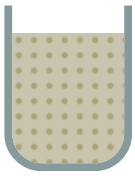
You may keep using A. Do not request the answer key yet. The images are model data, not direct product tests.

These images are static data from the current teaching model. Appearance answers are not filled in. They are not direct product-test results.

Control group: add 2 mL of water



Add 2 mL of digestive juice X



Add 2 mL of digestive juice Y



Add 1 mL each of digestive juices X and Y



Setup / conditions	Your appearance record (one per set)	
Control group: add 2 mL of water	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear
Add 2 mL of digestive juice X	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear
Add 2 mL of digestive juice Y	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear
Add 1 mL each of digestive juices X and Y	☐ Separate oil and water layers (no clear change you can see) ☐ become cloudy	☐ Separate oil and water layers (thinner oil layer) ☐ become clear

B3.2 X: choose the appearance, then explain it using emulsification (乳化作用).

Digestive juice X mainly causes lipids to form （a clear solution ／ a cloudy emulsion (乳狀液)）.

Brief reason / explanation:

B3.3 Total surface area after droplets become smaller: choose the answer and explain how this helps Y act.

Digestive juice X first breaks up large oil droplets into small droplets, causing their total surface area to （decrease ／ increase）, which helps digestive juice Y act.

Brief reason / explanation:

Complete and keep the first answers in B. After receiving C, correct in another colour without erasing the original.

Independent application: heating and evidence

Boil X or Y only. Cool to 37°C, keeping starting and total added volumes, time and shaking equal.

B4.1 Limits of appearance evidence: choose the best statement, then name another kind of evidence needed.

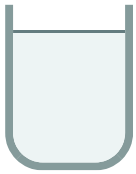
We only observed the oil layer and whether the mixture was cloudy or clear. We did not test for digestion products. Which statement fits best?（Appearance provides clues about the effects of digestive juices, but alone cannot prove that glycerol (甘油) and fatty acids (脂肪酸) have formed. ／ A clear mixture completely proves that lipids have broken down into glycerol (甘油) and fatty acids (脂肪酸).）

Brief reason / explanation:

B4.2 To study boiling one digestive juice, which set should be the baseline? What must happen to the boiled liquid before comparison?

B4.3 Record the appearances of the two boiled-and-cooled sets and compare them with unboiled X + Y.

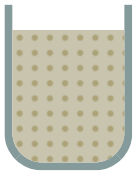
Add 1 mL each of digestive juices X and Y



Add 1 mL each of digestive juices X and Y (boil X, then cool)



Add 1 mL each of digestive juices X and Y (boil Y, then cool)



Setup / conditions	Your appearance record (one per set)	
Add 1 mL each of digestive juices X and Y (boil X, then cool)	☐ become cloudy	☐ become clear
Add 1 mL each of digestive juices X and Y (boil Y, then cool)	☐ become cloudy	☐ become clear

B4.4 Juice that loses its action after heating: choose the answer and explain why cooling does not restore its action.

Extension conclusion: the digestive juice that loses its function after boiling is（Digestive juice X ／ Digestive juice Y）.

Brief reason / explanation:

B4.5 Use at least one main-set comparison and one heating comparison to explain emulsification (乳化作用), enzyme action and denaturation. Name the sets. Do not claim a measured combined effect or increase in speed.

Complete and keep the first answers in B. After receiving C, correct in another colour without erasing the original.

Name: _____ Class and student number: _____

Check answers and correct

Give out C only after B is completed. Check, keep the first answers and correct them. Do not invent an original hypothesis.

Item	Check points; other sound wording is accepted
B1.1	Oil is above the water, with a clear boundary. Keep the visible description separate from a chemical explanation.
B1.2	The prediction can be tested, but the reason is wrong. Y is lipase (脂肪酶), which speeds up hydrolysis. Bile (膽汁) causes emulsification (乳化作用). Testable does not mean already proved.
B1.3	Independent: combination of juices. Dependent: appearance. Controlled: starting oil/water volume, temperature, time, total added volume and shaking method.
B1.4	Choose equal time, equal starting and total added volumes, and equal shaking; do not choose extra total liquid for X + Y.
B2.1	The 2 mL water set is the baseline for this question. Equal added volume avoids a difference in total volume or dilution.
B2.2	In order: water control, 2 mL water; X, 2 mL; Y, 2 mL; X + Y, 1 mL each.
B2.3	Unfair: 2 mL versus 4 mL. Use 1 mL of each juice, keeping the total at 2 mL to avoid a volume/dilution difference.
B3.1	Water: separated layers; X: cloudy; Y: thinner oil layer, still separated; X + Y: clear.
B3.2	Cloudy emulsion (乳狀液). X disperses oil droplets by emulsification (乳化作用), a form of physical digestion (物理消化), without changing the lipid molecules.
B3.3	Increases. For the same total amount of lipid, small droplets provide more surface for lipase (脂肪酶) to act on. This is not a measurement of speed.
B4.1	Choose the statement that appearance gives clues but does not prove glycerol (甘油) and fatty acids (脂肪酸) formed. Other chemical product tests and repeated tests are needed.
B4.2	The unboiled X + Y set, 1 mL each; cool the boiled juice to the common 37°C and keep the other conditions equal. This control contains X and Y.
B4.3	Boiled X + unboiled Y: clear; unboiled X + boiled Y: cloudy.
B4.4	Y. Lipase (脂肪酶) is a protein; boiling causes denaturation and loss of function. Cooling does not restore it in this model.
B4.5	For example, compare water with X for emulsification (乳化作用), and unboiled X + Y with boiled Y for enzyme loss. Explain bile (膽汁), lipase (脂肪酶) hydrolysis and heat denaturation. Appearance and different individual juice amounts limit measured claims. Other sound specific comparisons are accepted.

Correction 1

Choose an answer in B to correct. Write its ID, the sound explanation and why it was wrong. If all are correct, add evidence and a limit to one answer; do not invent a mistake.

Correction 2

Use one named comparison to explain emulsification (乳化作用), enzyme action or denaturation in your own words.

Appearance is a teaching clue: clarity alone does not prove glycerol (甘油) and fatty acids (脂肪酸) formed or measure speed. X + Y uses 1 mL each, while a single-juice set uses 2 mL. The different individual amounts cannot prove a measured combined effect or faster rate.

For teachers only. Do not give this part to students.

Research conditions, distribution and timing

This group uses the worksheet without operating the VL or receiving a lesson beforehand. Both groups have 15–20 minutes in total. Use the same T for both groups in each session, including self-study, practice and checking.

Choose a common T = _____ minutes (15–20) for both groups. Select it for the class reading pace and record actual start/end times. Checking is included; do not add a lesson beforehand.

Stage	T	T=15	T=20
A: explanations and examples	30%	4.5 min	6 min
B1–B2: guided practice	35%	5.25 min	7 min
B3–B4: independent application	25%	3.75 min	5 min
C: check and correct	10%	1.5 min	2 min

Distribution and teacher support

1. Give out A and B at the start. Keep A available. Explain only reading, answering and timing, not digestion concepts.
2. In B1–B2, clarify wording/procedure and point to A, without giving answers or deciding for students. Log extra clarification and use the common support rules for both groups.
3. B3–B4 are independent, with A available. Address procedure/access needs only, without answer feedback.
4. Save the first B answers before distributing C at checking time. Correct in another colour. Give brief consistent feedback using C, keeping first and corrected answers separate.
5. Give students only A–C. D and the complete pack are for teacher printing. Verify comparable actual total time from session records, not just a screen label.

Draft worksheet feedback guide. These marks are not equivalent to VL activity marks, SPSAI or CKT. Use separate common tests to assess research outcomes.

This material follows the specified self-study comparison condition; its activity process differs from IBL.

For teachers only. Do not give this part to students.

Full answer key and teaching limits

B1.1 • Describe only the oil–water test tube before any digestive juice is added. What layers and boundary can you see?

Oil is above the water, with a clear boundary. Keep the visible description separate from a chemical explanation.

B1.2 • Evaluate this given prediction and reason. You do not need to invent an “original hypothesis”.

The prediction can be tested, but the reason is wrong. Y is lipase (脂肪酶), which speeds up hydrolysis. Bile (膽汁) causes emulsification (乳化作用). Testable does not mean already proved.

B1.3 • The main comparison changes the combination of digestive juices. Classify the factors as independent, dependent or controlled variables.

Independent: combination of juices. Dependent: appearance. Controlled: starting oil/water volume, temperature, time, total added volume and shaking method.

B1.4 • Choose all the conditions needed for a fair comparison.

Choose equal time, equal starting and total added volumes, and equal shaking; do not choose extra total liquid for X + Y.

B2.1 • To compare adding digestive juice with adding only water, which of the four sets is the baseline? Why keep the total volume added the same?

The 2 mL water set is the baseline for this question. Equal added volume avoids a difference in total volume or dilution.

B2.2 • The four basic setups are already specified. Use the recipes in A to complete each label and added volume; no new apparatus design is needed.

In order: water control, 2 mL water; X, 2 mL; Y, 2 mL; X + Y, 1 mL each.

B2.3 • Someone adds 2 mL water to the control but 2 mL X and 2 mL Y to the combined set, with other conditions equal. Is this fair? Identify the problem, fix the two volumes and explain why.

Unfair: 2 mL versus 4 mL. Use 1 mL of each juice, keeping the total at 2 mL to avoid a volume/dilution difference.

Appearance is a teaching clue: clarity alone does not prove glycerol (甘油) and fatty acids (脂肪酸) formed or measure speed. X + Y uses 1 mL each, while a single-juice set uses 2 mL. The different individual amounts cannot prove a measured combined effect or faster rate.

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Full answer key and teaching limits

B3.1 • Independent application: observe the four static images and record their appearances.

Water: separated layers; X: cloudy; Y: thinner oil layer, still separated; X + Y: clear.

B3.2 • X: choose the appearance, then explain it using emulsification (乳化作用).

Cloudy emulsion (乳状液). X disperses oil droplets by emulsification (乳化作用), a form of physical digestion (物理消化), without changing the lipid molecules.

B3.3 • Total surface area after droplets become smaller: choose the answer and explain how this helps Y act.

Increases. For the same total amount of lipid, small droplets provide more surface for lipase (脂肪酶) to act on. This is not a measurement of speed.

B4.1 • Limits of appearance evidence: choose the best statement, then name another kind of evidence needed.

Choose the statement that appearance gives clues but does not prove glycerol (甘油) and fatty acids (脂肪酸) formed. Other chemical product tests and repeated tests are needed.

B4.2 • To study boiling one digestive juice, which set should be the baseline? What must happen to the boiled liquid before comparison?

The unboiled X + Y set, 1 mL each; cool the boiled juice to the common 37°C and keep the other conditions equal. This control contains X and Y.

B4.3 • Record the appearances of the two boiled-and-cooled sets and compare them with unboiled X + Y.

Boiled X + unboiled Y: clear; unboiled X + boiled Y: cloudy.

B4.4 • Juice that loses its action after heating: choose the answer and explain why cooling does not restore its action.

Y. Lipase (脂肪酶) is a protein; boiling causes denaturation and loss of function. Cooling does not restore it in this model.

B4.5 • Use at least one main-set comparison and one heating comparison to explain emulsification (乳化作用), enzyme action and denaturation. Name the sets. Do not claim a measured combined effect or increase in speed.

For example, compare water with X for emulsification (乳化作用), and unboiled X + Y with boiled Y for enzyme loss. Explain bile (膽汁), lipase (脂肪酶) hydrolysis and heat denaturation. Appearance and different individual juice amounts limit measured claims. Other sound specific comparisons are accepted.

Appearance is a teaching clue: clarity alone does not prove glycerol (甘油) and fatty acids (脂肪酸) formed or measure speed. X + Y uses 1 mL each, while a single-juice set uses 2 mL. The different individual amounts cannot prove a measured combined effect or faster rate.

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Question-by-question mapping

Original VL item	Treatment	Revised item	Shared goal	Reason
Your first observation： Describe only what you can see in the oil + water test tube (do not yet guess what X and Y are or what the results will be).	Retained / equivalent rewrite	B1.1 Describe only the oil–water test tube before any digestive juice is added. What layers and boundary can you see?	Observation: separate what is seen from an explanation.	Keep the common content and skills, practised through self-study or static data.
Complete your prediction / If I add _____to the oil and water, the mixture in the test tube will _____. / My reason (required; there is no single correct answer)	Rewritten	B1.2 Evaluate this given prediction and reason. You do not need to invent an “original hypothesis”.	Inquiry design: judge whether a prediction is testable and whether its reason is sound.	Teach the method with an example, then evaluate another prediction and reason.
02 • Fair test design： Independent variable (the factor changed on purpose) / Dependent variable (the result you measure or observe) / Controlled variables (the factors kept the same)	Retained / equivalent rewrite	B1.3 The main comparison changes the combination of digestive juices. Classify the factors as independent, dependent or controlled variables.	Classification: identify the three kinds of variables.	Keep the common content and skills, practised through self-study or static data.
Which conditions does this fair test need?	Retained / equivalent rewrite	B1.4 Choose all the conditions needed for a fair comparison.	Inquiry design and experimental reasoning: identify sound fixed conditions.	Keep the common content and skills, practised through self-study or static data.
Does this experiment need a control group? If so, how would you design the setups to compare?	Rewritten	B2.1 To compare adding digestive juice with adding only water, which of the four sets is the baseline? Why keep the total volume added the same?	Inquiry design: explain the purpose of a control.	Keep the common content and skills, practised through self-study or static data.

For teachers only. Do not give this part to students.

Question-by-question mapping

Original VL item	Treatment	Revised item	Shared goal	Reason
Written design (can replace a drawing): describe the four setups, how much liquid you add and which conditions you keep the same. / Draw and label your setup / Use your experimental variables to help you draw and label your setup.	Rewritten	B2.2 The four basic setups are already specified. Use the recipes in A to complete each label and added volume; no new apparatus design is needed.	Experimental reasoning and communication: read and label setups and volumes.	Label and evaluate a given procedure while practising apparatus and fair-comparison skills.
Written design (can replace a drawing): describe the four setups, how much liquid you add and which conditions you keep the same. / Does this experiment need a control group? If so, how would you design the setups to compare?	Rewritten	B2.3 Someone adds 2 mL water to the control but 2 mL X and 2 mL Y to the combined set, with other conditions equal. Is this fair? Identify the problem, fix the two volumes and explain why.	Inquiry design and experimental reasoning: evaluate and correct an unfair procedure.	Keep the common content and skills, practised through self-study or static data.
Select the appearance based on what you observe:	Static data	B3.1 Independent application: observe the four static images and record their appearances.	Observation and communication: distinguish the four appearances.	Keep the common content and skills, practised through self-study or static data.
1. What does digestive juice X do?：Digestive juice X mainly causes lipids to form _____.	Retained / equivalent rewrite	B3.2 X: choose the appearance, then explain it using emulsification (乳化作用).	Inference: link an appearance with emulsification (乳化作用).	Keep the common content and skills, practised through self-study or static data.
2. How does digestive juice X help digestive juice Y act?：Digestive juice X first breaks up large oil droplets into small droplets, causing their total surface area to _____, which helps digestive juice Y act.	Retained / equivalent rewrite	B3.3 Total surface area after droplets become smaller: choose the answer and explain how this helps Y act.	Inference: explain surface area without inventing speed data.	Keep the common content and skills, practised through self-study or static data.

For teachers only. Do not give this part to students.

Question-by-question mapping

Original VL item	Treatment	Revised item	Shared goal	Reason
3. What can this experiment tell us?：We only observed the oil layer and whether the mixture was cloudy or clear. We did not test for digestion products. Which statement fits best?_____	Retained / equivalent rewrite	B4.1 Limits of appearance evidence: choose the best statement, then name another kind of evidence needed.	Inference and communication: distinguish clues, direct evidence and limits.	Keep the common content and skills, practised through self-study or static data.
Does this experiment need a control group? If so, how would you design the setups to compare? / Which digestive juice is affected by high temperature?	Rewritten	B4.2 To study boiling one digestive juice, which set should be the baseline? What must happen to the boiled liquid before comparison?	Inquiry design and experimental reasoning: choose the control for the question and control reaction temperature.	Keep the common content and skills, practised through self-study or static data.
Use “add 1 mL each of digestive juices X and Y” as the group for comparison. Boil only one digestive juice at a time. Complete both tests and record your observations.	Static data	B4.3 Record the appearances of the two boiled-and-cooled sets and compare them with unboiled X + Y.	Observation: compare heating treatments with a baseline.	Keep the common content and skills, practised through self-study or static data.
Extension conclusion: the digestive juice that loses its function after boiling is_____.	Retained / equivalent rewrite	B4.4 Juice that loses its action after heating: choose the answer and explain why cooling does not restore its action.	Inference and content knowledge: explain heat treatment using enzyme denaturation.	Keep the common content and skills, practised through self-study or static data.
Do your observations support your first idea to test? Name the groups you compared and what you saw. Use the key learning points to improve your explanation.	Rewritten	B4.5 Use at least one main-set comparison and one heating comparison to explain emulsification (乳化作用), enzyme action and denaturation. Name the sets. Do not claim a measured combined effect or increase in speed.	Communication and content knowledge: explain concepts using named comparisons.	Concepts are taught first; assess their use with evidence rather than inventing an original hypothesis.

For teachers only. Do not give this part to students.

Worksheet feedback guide

Draft worksheet feedback guide. These marks are not equivalent to VL activity marks, SPSAI or CKT. Use separate common tests to assess research outcomes.

Item	Maximum	Shared skill / knowledge	Marking criteria
B1.1	2	Observation: separate what is seen from an explanation.	2: describes both layers and the boundary without claiming digestion; 1: one specific visible feature; 0: no relevant description.
B1.2	2	Inquiry design: judge whether a prediction is testable and whether its reason is sound.	2: separates testability from the wrong reason and explains enzyme action versus emulsification (乳化作用); 1: one useful judgement or explanation; 0: neither.
B1.3	4	Classification: identify the three kinds of variables.	Independent 1, dependent 1; five controlled factors at 0.4 each, total 2. A factor from a wrong category gives 0 for that category.
B1.4	1	Inquiry design and experimental reasoning: identify sound fixed conditions.	1 for all three correct conditions with no wrong choice; 0 for another attempted set.
B2.1	2	Inquiry design: explain the purpose of a control.	1 for the baseline, 1 for a sound reason to keep total volume fixed.
B2.2	2	Experimental reasoning and communication: read and label setups and volumes.	2 for all four correct setup/volume labels; 0.5 per complete label.
B2.3	3	Inquiry design and experimental reasoning: evaluate and correct an unfair procedure.	1 for the flaw, 1 for corrected volumes, 1 for the fairness explanation.
B3.1	4	Observation and communication: distinguish the four appearances.	1 per correctly described set, total 4.

B first-answer practice maximum: 36. Use C corrections for separate feedback; do not overwrite the first-answer marks.

Mark missing answers as unavailable and follow the study’ s missing-data rule. Do not invent evidence. Judge explanations and links to data, not keywords.

These marks are not equivalent to VL original-prediction, virtual-operation or completion marks and must not be converted to its 32-point score. Assess research outcomes using separate common tests.

For teachers only. Do not give this part to students.

Worksheet feedback guide

Draft worksheet feedback guide. These marks are not equivalent to VL activity marks, SPSAI or CKT. Use separate common tests to assess research outcomes.

Item	Maximum	Shared skill / knowledge	Marking criteria
B3.2	2	Inference: link an appearance with emulsification (乳化作用).	1 for the appearance; 1 for a sound explanation of emulsification (乳化作用) without molecular change.
B3.3	2	Inference: explain surface area without inventing speed data.	1 for increase, 1 for explaining the available surface and enzyme action.
B4.1	2	Inference and communication: distinguish clues, direct evidence and limits.	1 for the sound statement, 1 for a useful additional test or repeated evidence.
B4.2	2	Inquiry design and experimental reasoning: choose the control for the question and control reaction temperature.	1 for the unboiled baseline, 1 for cooling to the common temperature for a fair comparison.
B4.3	2	Observation: compare heating treatments with a baseline.	1 for each set.
B4.4	2	Inference and content knowledge: explain heat treatment using enzyme denaturation.	1 for Y, 1 for protein-enzyme denaturation and why cooling alone does not restore it.
B4.5	4	Communication and content knowledge: explain concepts using named comparisons.	1 each for emulsification (乳化作用), enzyme hydrolysis and denaturation; 1 for clear main/heating comparisons. Judge explanations, not keywords.

B first-answer practice maximum: 36. Use C corrections for separate feedback; do not overwrite the first-answer marks.

Mark missing answers as unavailable and follow the study’ s missing-data rule. Do not invent evidence. Judge explanations and links to data, not keywords.

These marks are not equivalent to VL original-prediction, virtual-operation or completion marks and must not be converted to its 32-point score. Assess research outcomes using separate common tests.